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Last time :

$$S = \{n \geq 1 : n = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}\}.$$

Thm. Given an integer $n \geq 1$, $n \in S$ if and only if
for every $p|n$ with $p \equiv 3 \pmod{4}$, $e_p(n)$ is even.

Necessity ✓

Prop. S is multiplicatively closed : $m, n \in S \Rightarrow mn \in S$.

Thm (Fermat - Euler) Every prime $p \equiv 1 \pmod{4}$ can be written
as a sum of two squares.

Pf of the sufficiency part: $1 \in S$.

Suppose $n = p_1 p_2 \cdots p_k g_1^{e_1} g_2^{e_2} \cdots g_t^{e_t} > 1$, where

each of $p_1, \dots, p_k$ is either 2 or a prime $\equiv 1 \pmod{4}$,

each of $g_1, \dots, g_t$ is a prime $\equiv 3 \pmod{4}$ with an even exponent e_j .

Note that $2 = 1^2 + 1^2 \in S$ and Fermat-Euler $\Rightarrow$ odd $p_i \in S$

Since e_j is even, $g_j^{e_j} = (g_j^{\frac{e_j}{2}})^2 \in S$.

By Prop, $n \in S$.

□

Note: If $n = x^2 + y^2$ with $x \geq y$, then $x^2 \leq n$ and

$n = x^2 + y^2 \leq x^2 + x^2 = 2x^2$. So $\sqrt{\frac{n}{2}} \leq x \leq \sqrt{n}$. Similarly,

$$y \leq \sqrt{\frac{n}{2}}.$$

Unique factorization beyond $\mathbb{Z}$.

Let D be an integral domain, which is a commutative ring with 1 s.t. $ab=0 \Rightarrow a=0$ or $b=0$.

Defn (Divisibility) For $\alpha, \beta \in D$ with $\alpha \neq 0$, we say α divides β and write $\alpha | \beta$ if there is $r \in D$ s.t. $\beta = \alpha r$.

Defn (Unit) A unit $u \in D$ is a nonzero element s.t. there is $v \in D$ s.t. $uv = vu = 1$.

Properties of divisibility

1. Reflexivity: $\alpha | \alpha$

2. Anti-symmetry: $\alpha | \beta, \beta | \alpha \Rightarrow \beta = u\alpha$ for some unit $u \in D$

Pf. $\alpha/\beta \Rightarrow \beta = u\alpha$ for some $u \in D$.

$\beta/\alpha \Rightarrow \alpha = v\beta$ for some $v \in D$.

$$\text{So } \beta = u\alpha = u(v\beta) = (uv)\beta.$$

Since $\beta \neq 0$, $1 = uv$. So u is a unit. □

3. Transitivity: $\alpha/\beta, \beta/\gamma \Rightarrow \alpha/\gamma$

4. Linearity: $\alpha/\beta, \alpha/\gamma \Rightarrow \alpha/x\beta + y\gamma$ for any $x, y \in D$

5. Size: α/β ($\beta \neq 0$) $\Rightarrow \alpha$ is "smaller" than β ?

Defn (Associate) An associate of α is of the form $u\alpha$ with a unit $u \in D$.

Defn (Primes and composites) We say that a nonzero nonunit

$\alpha \in D$ is a prime if its only divisors are units and its associates. Otherwise, we say it is composite.

Recall the proof of FTA:

size $\Rightarrow$ existence

DA $\Rightarrow \mathbb{Z}$ is a PID $\stackrel{\text{gcd}}{\Rightarrow}$ Bezout's lemma

$\Rightarrow$ Euclid's lemma

$\Rightarrow$ uniqueness

The ring $\mathbb{Z}[\sqrt{d}]$

Let $d \in \mathbb{Z}$. Define

$$\mathbb{Z}[\sqrt{d}] = \{x + y\sqrt{d} : x, y \in \mathbb{Z}\}.$$

When $d = -1$, a complex number $x + y\sqrt{-1} = x + yi$ with

$x, y \in \mathbb{Z}$ is called a Gaussian integer. The set of all Gaussian integers is denoted by

$$\mathbb{Z}[i] = \{x + yi : x, y \in \mathbb{Z}\}.$$

Note: ① $\mathbb{Z} \subseteq \mathbb{Z}[\sqrt{d}] \subseteq \mathbb{C}$.

② $\mathbb{Z}[\sqrt{d}]$ is an integral domain under $+$ and $\times$:

$$(x_1 + y_1\sqrt{d}) + (x_2 + y_2\sqrt{d}) = (x_1 + x_2) + (y_1 + y_2)\sqrt{d}.$$

$$(x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d}) = (x_1x_2 + y_1y_2d) + (x_1y_2 + x_2y_1)\sqrt{d}.$$

Defn (Norm) For $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$, the norm of α is defined as

$$N(\alpha) = (x + y\sqrt{d})(x - y\sqrt{d}) = x^2 - dy^2 \in \mathbb{Z}.$$

Defn (Conjugate) The conjugate of $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ is

$$\bar{\alpha} = x - y\sqrt{d}.$$

$$\text{So } N(\alpha) = N(\bar{\alpha}) = \alpha\bar{\alpha}.$$

From now on, we will assume that d is not a square,

so that $\mathbb{Z}[\sqrt{d}]$ strictly contains $\mathbb{Z}$.

Observation: If d is not a square, then

$$N(\alpha) = 0 \Leftrightarrow \alpha = 0.$$

$$\alpha = x + y\sqrt{d} = 0 \Rightarrow x = -y\sqrt{d}$$

$$\Rightarrow x^2 = y^2 d$$

$$\Rightarrow N(\alpha) = x^2 - y^2 d = 0.$$

$$N(\alpha) = 0 \Rightarrow x^2 - y^2 d = 0 \Rightarrow x^2 = y^2 d$$